Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

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Abstract

We present Fast3R, a transformer-based architecture for multi-view 3D reconstruction that processes 1000+ unordered, unposed images in a single forward pass. *Existing state-of-the-art methods such as DUSt3R operate pairwise, requiring $O(N^2)$ image pairs and a costly global alignment step that becomes a bottleneck as the number of views grows. Fast3R eliminates this bottleneck by performing all-to-all attention over patches from every input view simultaneously, with image-index positional embeddings to distinguish views. A key design choice is* random image-index positional embedding sampling *during training, which enables generalization from 20 training views to 1000+ views at inference—a form of "train short, test long." Dual DPT decoder heads regress both local and global 3D pointmaps with per-pixel confidence, and a local-to-global alignment strategy improves reconstruction fidelity. On camera pose estimation (CO3Dv2, 10 views), Fast3R achieves RRA@15 of 99.7% at 251.1 FPS—over 300× faster than DUSt3R (0.78 FPS)—while matching or surpassing it in accuracy. On scene reconstruction benchmarks (7-Scenes and NRGBD), Fast3R is competitive with DUSt3R at a fraction of the computational cost. At 32 views on a single A100, Fast3R runs in 0.51 s versus DUSt3R's 129 s; DUSt3R runs out of memory beyond 48 views, while Fast3R scales to 1500 views.*

1. Introduction

Recovering 3D structure from multiple posed or unposed images is a fundamental problem in computer vision, with applications ranging from robotics and augmented reality to cultural heritage preservation. Classical structure-from-motion (SfM) pipelines [21, 22] achieve impressive accuracy but rely on hand-crafted feature extractors [6, 20], global bundle adjustment, and hours of computation for large-scale scenes.

Recent learning-based approaches have dramatically simplified this pipeline. DUSt3R [33] introduced the idea of regressing 3D pointmaps directly from image pairs with a transformer, eliminating the need for explicit feature matching and camera calibration. MASt3R [10] extended this with richer local feature descriptors. However, both methods are fundamentally *pairwise*: to handle N images, they enumerate $O(N^2)$ pairs and then run a global alignment step to reconcile the pairwise predictions into a consistent world-frame reconstruction. This approach suffers from quadratic scaling in computation, memory proportional to the number of pairs, error accumulation across the alignment, and prohibitive latency for large view counts.

Spann3R [29] proposed processing views sequentially with an external memory, reducing the number of passes, but the sequential nature limits parallelism and scalability.

In this paper, we propose Fast3R, which reframes multi-view 3D reconstruction as a *single forward pass* over all N views simultaneously. The key idea is to concatenate all image patch tokens along the sequence dimension and run a transformer with all-to-all attention, allowing the model to globally reconcile geometry across views without any post-hoc alignment. Two prediction heads—one for local and one for global pointmaps—produce dense per-pixel 3D point predictions with confidence scores for every view jointly.

The central challenge in training such a model is view-count generalization: it is impractical to train with 1000 views, yet the model must handle such counts at inference. We address this with *random image-index positional embedding sampling*: during training, we randomly sample image indices from a large pre-allocated sinusoidal embedding table (up to index 999), even when training with only 20 views. This means that the model never sees the same ordering or the sequential indices 0–19 and thus learns to treat image indices as arbitrary identifiers rather than sequential positions, enabling smooth extrapolation to arbitrary view counts at test time.

Our contributions are as follows:

- A single-pass multi-view pointmap regressor with all-to-all attention over all N views, eliminating $O(N^2)$ pairwise inference and global alignment.
- Image-index sinusoidal positional embeddings with random index sampling during training, enabling generaliza-

tion from 20 training views to 1000+ inference views.

- A dual-head design (local and global DPT heads) with confidence-weighted supervision; using the local pointmap aligned to the global frame improves reconstruction fidelity.
- Extensive experiments demonstrating state-of-the-art pose accuracy on CO3Dv2 (RRA@15=99.7%), competitive reconstruction on 7-Scenes and NRGBD, and unprecedented scalability to 1000+ views on a single GPU.

2. Related Work

Classical Multi-View Reconstruction. Traditional SfM pipelines [21, 22] detect and match local features [6, 7, 14, 20, 38] across pairs of images, estimate relative poses, and then perform bundle adjustment to jointly refine all camera parameters and a sparse point cloud. While these pipelines achieve excellent accuracy, they require careful tuning of many stages and scale poorly to large scenes. Multi-view stereo methods [9, 36] produce dense reconstructions but also require known camera poses as input.

Learning-Based Pose Estimation. Several recent works learn to predict relative camera poses directly from image pairs. RelPose [40] and PoseDiffusion [31] use neural networks to regress pairwise rotations and translations. RelPose++ [13] improves on these with better training. These methods still require a separate global registration step to combine pairwise predictions into a consistent multi-view solution. VGGSfM [30] and RayDiffusion [41] propose end-to-end differentiable SfM pipelines with learned components.

Dense Pointmap Regression. DUSt3R [32, 33] introduced the paradigm of regressing 3D pointmaps—dense per-pixel 3D coordinate maps—from image pairs using a CroCo [34]-pretrained transformer. By expressing geometry as pointmaps, DUSt3R avoids explicit camera parameter estimation and achieves strong generalization. A global alignment step reconciles pairwise predictions. MASt3R [10, 11] augments DUSt3R with dense local feature maps for improved matching. MonST3R [39] extends to monocular video with dynamic scenes.

Multi-View Feed-Forward Reconstruction. Spann3R [29] processes views one at a time, maintaining a spatial memory of past observations. This avoids pairwise enumeration but is inherently sequential and cannot fully leverage global context. FlowMap [24] recovers camera poses from optical flow. InstantSplat [8] uses DUSt3R pointmaps to initialize 3D Gaussian splatting. In contrast, Fast3R processes all views simultaneously in a single forward pass.

Neural Rendering and Gaussian Splatting. Neural Radiance Fields (NeRF) [?] and 3D Gaussian Splatting (3DGS) [?] achieve photorealistic novel-view synthesis but require known camera poses, typically supplied by a classical SfM pipeline. Recent works such as InstantSplat [8] relax this requirement by initializing Gaussian splatting from DUSt3R pointmaps. Fast3R is a natural upstream component for such pipelines: its single-pass, pose-free pointmap estimates can directly seed downstream rendering approaches, enabling end-to-end scene reconstruction and rendering from completely unposed images at dramatically lower latency.

Visual SLAM. Simultaneous Localization and Mapping (SLAM) systems [16, 27] jointly estimate camera trajectories and sparse maps in real time, making them well-suited for robotics and augmented reality. Learning-based SLAM such as DROID-SLAM [26] leverages differentiable bundle adjustment and dense optical flow for robust pose estimation on video sequences. Unlike SLAM methods, which process frames incrementally and require temporal ordering, Fast3R processes all views simultaneously in a single forward pass with no assumed ordering or initialization, enabling both video and unordered image collections.

Transformers for Vision. Vision transformers [28] have become the backbone of choice for many vision tasks. DINOv2 [17] provides powerful pre-trained features. FlashAttention [4, 5] makes attention over long sequences practical. DPT [18] produces dense predictions from transformer features. RoPE [25] and position interpolation [3] allow length generalization. Fast3R leverages all of these advances.

3. Method

Fast3R takes as input a set of N unordered, unposed images $\{I_i\}_{i=1}^N$ and produces per-pixel 3D pointmaps $\{X_i^{\text{global}}, X_i^{\text{local}}\}_{i=1}^N$ with confidence maps $\{C_i^{\text{global}}, C_i^{\text{local}}\}_{i=1}^N$ in a single forward pass. Figure 1 shows the overall architecture.

3.1. Image Encoder

Each image is independently processed by a shared ViT-L [28] encoder (embed dim 1024, 16 heads, depth 24, patch size 16) with RoPE [25] 2D positional embeddings and FlashAttention [4, 5]. This is initialized from a CroCo [34]-pretrained checkpoint and fine-tuned end-to-end. The encoder supports variable-resolution inputs via a multi-aspect-ratio patch embedding (ManyAR_PatchEmbed), allowing images of different sizes and aspect ratios within the same batch.

For image I_i, the encoder outputs a sequence of patch tokens $F_i \in \mathbb{R}^{B \times P_i \times D_e}$ and associated 2D spatial positions

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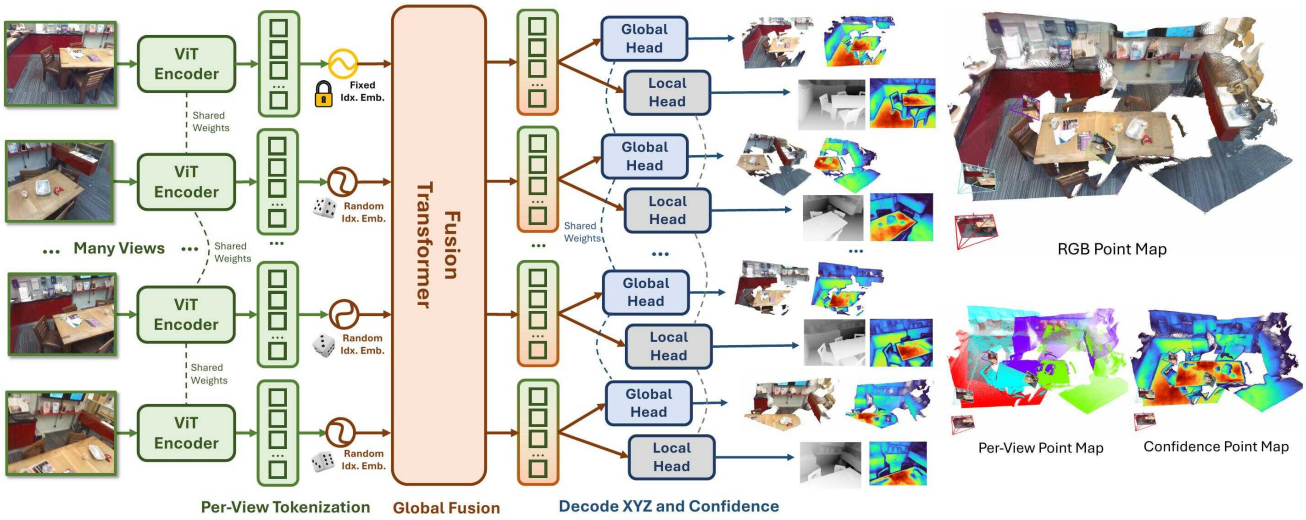

Figure 1. **Model architecture of Fast3R.** Given N unordered, unposed images, a shared ViT-L encoder independently extracts patch tokens from each view. The tokens from all views are concatenated and fed into a fusion transformer with all-to-all attention. Image-index positional embeddings distinguish patches from different views. Two DPT heads predict local and global pointmaps with per-pixel confidence scores. No global alignment or pairwise enumeration is required.

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pos_i, where P_i is the number of patches and $D_e = 1024$.

3.2. Fusion Transformer Decoder with Image-Index Positional Embeddings

The patch tokens from all N views are concatenated along the sequence dimension:

$$F = [F_1; F_2; \ldots; F_N] \in \mathbb{R}^{B \times P_{\text{total}} \times D_e}, \quad (1)$$

where $P_{\text{total}} = \sum_i P_i$. This concatenated sequence is then projected to the decoder dimension $D_d = 768$ via a linear layer.

To distinguish patches from different views, we add *image-index positional embeddings* (image-index PE). We maintain a fixed sinusoidal embedding table $E \in \mathbb{R}^{1000 \times D_d}$, and each patch from view i receives the embedding $E[\text{idx}_i]$, where idx_i is the index assigned to view i. These embeddings are added to the projected patch tokens before the fusion transformer.

Random Image-Index PE Sampling. A critical design choice for enabling "train short, test long" generalization is the sampling strategy for image indices. During training with N_{train} views (e.g., 20), instead of always assigning indices $\{0, 1, \ldots, N_{\text{train}} - 1\}$, we randomly sample N_{train} non-repeating indices from the full table of 1000 (with view 0 always assigned index 0, and the rest randomly permuted). This prevents the model from associating specific indices with positions in a sequence and instead learns to use image-index PE as arbitrary view identifiers. As a result, at test time with 1000 views, the model simply uses indices $\{0, 1, \ldots, 999\}$ from the pre-allocated table, and all embeddings are within the training distribution. Figure 8 validates this design choice.

The fusion transformer consists of 12 transformer blocks (embed dim 768, 12 heads) with all-to-all attention using FlashAttention and RoPE 2D positional embeddings over spatial patch coordinates. Each block can attend to all patches from all views, enabling global geometric reasoning without any view-specific masking.

3.3. Dual DPT Prediction Heads

We use two independent DPT [18] regression heads—one for global pointmaps and one for local pointmaps—each taking multi-scale features from the decoder and producing dense per-pixel 3D coordinates and confidence values.

The **global head** regresses pointmaps in a shared world coordinate frame, enabling consistent multi-view reconstruction without post-hoc alignment.

The **local head** regresses pointmaps in each view's own camera coordinate frame, producing view-centric geometry that is easier to supervise and more accurate in fine-grained local structure.

At inference, we use the local pointmaps aligned to the global coordinate frame (via the global head's reference) for final reconstruction, as this consistently outperforms using global pointmaps alone (Section 5).

To handle the large number of views efficiently, the DPT heads are applied in parallel chunks of up to 25 views at a time to avoid memory overflow.

3.4. Local-to-Global Alignment

At inference, we align each view's local pointmap into the global coordinate frame using a confidence-weighted Procrustes step. For each view i, we solve for a rigid transformation $T_i \in SE(3)$ that minimizes the confidence-weighted point-to-point distance:

$$T_i = \arg\min_{T \in SE(3)} \sum_j C_{i,j}^{\text{global}} \| T X_{i,j}^{\text{local}} - X_{i,j}^{\text{global}} \|_2^2, \quad (2)$$

which admits a closed-form solution via singular value decomposition. The final reconstruction uses the transformed local pointmaps $\{T_i X_i^{\text{local}}\}_{i=1}^N$ rather than the global pointmaps directly. Because the local head is supervised in per-view camera coordinates, it produces sharper, more accurate local geometry than the global head, which must simultaneously represent all views in a single shared frame. Aligning local predictions to the global reference therefore combines precise per-view geometric detail with globally consistent camera poses, yielding the reconstruction improvements reported in Section 5.

3.5. Training Objective

Following DUSt3R [33], we train with a confidence-weighted normalized pointmap loss. For each view i and each pixel j, the global loss is:

$$\mathcal{L}_{\text{global}} = \sum_{i,j} C_{i,j} \cdot \| X_{i,j}^{\text{global}} - \hat{X}_{i,j}^{\text{global}} \|_2 - \alpha \log C_{i,j}, \quad (3)$$

where $C_{i,j}$ is the predicted confidence, $\hat{X}$ denotes ground-truth pointmaps (normalized by their average distance to the origin), and $\alpha = 0.2$ balances the confidence regularization. An analogous loss is applied to the local head. The total loss is the sum of both.

We optimize with AdamW [15] (lr $= 10^{-4}$, $\beta = (0.9, 0.95)$, weight decay 0.05) and a linear warmup cosine annealing scheduler.

3.6. Implementation Details

Fast3R is initialized from a DUSt3R [33] checkpoint and fine-tuned end-to-end in two stages. The first stage trains with $N = 2$ views for 50K steps using the same training mixture as DUSt3R, providing a strong geometric prior. The second stage trains with up to $N = 20$ views for 20K steps with random image-index PE sampling enabled, teaching the model global multi-view reasoning. Both stages use AdamW with a linear warmup cosine decay schedule on 32 A100 GPUs with a total batch size of 128 and mixed-precision (BF16) arithmetic. Training data includes BlendedMVS [37], MegaDepth [12], and the outdoor scene datasets used by DUSt3R [33], providing diverse coverage of object-centric and scene-level geometry across varied lighting conditions and scales. Images are

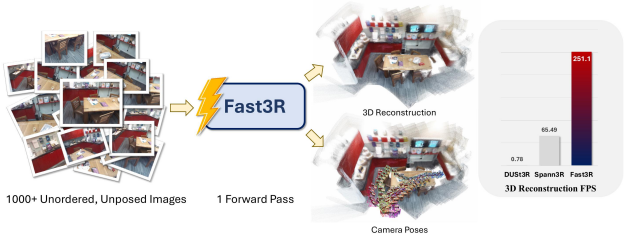

Figure 2. **Fast3R** reconstructs 1000+ unordered, unposed images in a single forward pass, achieving orders-of-magnitude speedups over DUSt3R while maintaining competitive accuracy.

resized to fit within a 512×512 pixel budget while preserving aspect ratio, with multiple aspect ratios supported within the same batch via the multi-aspect-ratio patch embedding. At inference, camera poses are recovered from predicted global pointmaps via confidence-weighted singular value decomposition, requiring no iterative refinement or post-processing steps, which contributes to Fast3R's low end-to-end latency.

3.7. Scalability

The key to Fast3R's scalability is the elimination of $O(N^2)$ pairwise inference and global alignment. The transformer attention over all patches is $O(P_{\text{total}}^2)$ in sequence length, but with FlashAttention, memory and time scale linearly with sequence length in practice. Furthermore, Fast3R is trivially parallelizable across GPUs along the view dimension for both encoding and the DPT heads, enabling deployment at scale.

Beyond single-GPU deployment, Fast3R's architecture is well-suited for multi-GPU inference. Image encoding is fully independent across views, and the DPT heads operate on per-view chunks, so both stages parallelize across devices with minimal communication overhead. The fusion transformer can additionally be sharded using sequence-parallel strategies, distributing the attention computation across devices. In practice, we observe near-linear speedups when spreading views across multiple A100 GPUs, suggesting that Fast3R can scale to arbitrarily large image collections given sufficient hardware. This property stands in sharp contrast to pairwise methods such as DUSt3R and MASt3R, whose $O(N^2)$ inference structure makes cross-device load balancing fundamentally more challenging.

4. Experiments

We evaluate Fast3R on camera pose estimation and 3D reconstruction benchmarks, and analyze its computational scaling behavior.

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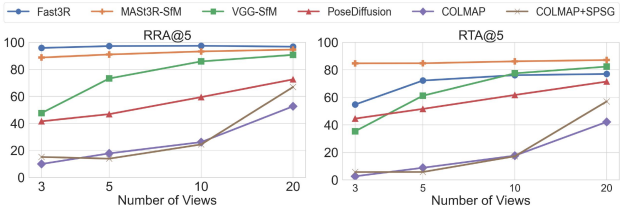

Figure 3. **Pose accuracy with more views:** Fast3R improves with the context from more views, demonstrating positive scaling with view count.

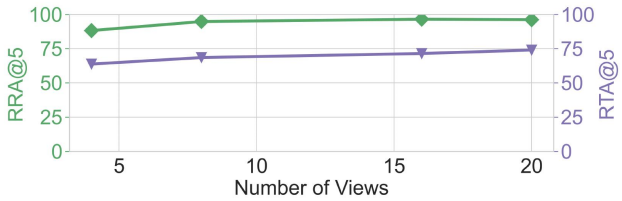

Figure 4. **Increasing # views during training: camera pose estimation on CO3D.** Training with more views per scene consistently improves pose accuracy.

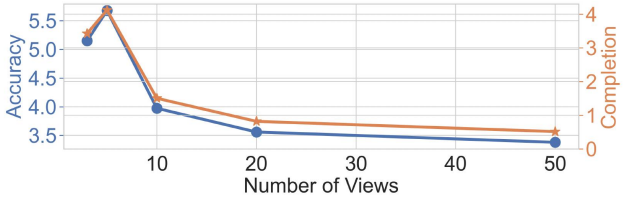

Figure 5. **DTU reconstruction quality vs. test number of views.** Fast3R improves monotonically with more input views.

4.1. Camera Pose Estimation

Datasets and Metrics. We evaluate on CO3Dv2 [19] (object-centric, 10 views per scene) and RealEstate10K [42] (indoor scene video). We report Rotation/Translation Accuracy at threshold τ degrees (RRA@τ, RTA@τ) and mean Average Accuracy mAA(30). Fast3R does not assume known camera intrinsics.

Baselines. We compare against COLMAP+SuperPoint+SuperGlue [6, 20, 21], PixSfM [14], RelPose [40], PoseDiffusion [31], RelPose++ [13], DUSt3R [33], and MASt3R [10].

Results. Table 1 shows results on CO3Dv2 and RealEstate10K. Fast3R-no-outdoor (trained without outdoor data, matching CO3Dv2 domain) achieves RRA@15=99.7%, RRA@5=97.4%, and mAA(30)=82.5 on CO3Dv2—outperforming all baselines including MASt3R. Fast3R (trained with all data) achieves mAA(30)=72.7 on RealEstate10K, demonstrating competitive cross-domain generalization.

Crucially, Fast3R runs at 251.1 FPS, more than 300× faster than DUSt3R (0.78 FPS) and over 1000× faster than MASt3R (0.23 FPS).

Scaling with Number of Views. Figure 3 shows that Fast3R's pose accuracy *improves* as more views are provided at test time, demonstrating positive scaling behavior. Similarly, Figure 4 shows that training with more views per scene improves pose estimation accuracy on CO3D, highlighting the benefit of global multi-view reasoning.

4.2. 3D Reconstruction

Scene Datasets. We evaluate reconstruction quality on 7-Scenes [23] (indoor RGB-D, 500–1500 frames per scene) and NRGBD [2] (indoor video). We report median accuracy (Acc↓) and completeness (Comp↓) of reconstructed point clouds against ground truth, following the same frame skip as prior work (skip=20 for 7-Scenes, skip=40 for NRGBD). Values are scaled 100×.

Table 2 shows results. Fast3R achieves 1.58/0.93 (Acc/Comp) on 7-Scenes and 3.40/1.01 on NRGBD, at 251.1 FPS—substantially faster than DUSt3R (0.78 FPS) and Spann3R (65.4 FPS). DUSt3R slightly outperforms Fast3R in Acc (1.23 vs 1.58 on 7-Scenes) due to its global alignment, but Fast3R is within competitive range while being orders of magnitude faster.

Object-Centric Dataset. Table 3 shows results on DTU [1], an object-centric dataset with 49-frame trajectories (using skip=5). Fast3R achieves the best completeness (0.857) and competitive accuracy (1.706) compared to DUSt3R (1.159/0.914) and Spann3R (2.268/1.295).

Figure 5 shows DTU reconstruction quality as a function of the number of test views. Fast3R consistently improves with more views, again demonstrating positive scaling. Figure 6 shows that training with more views per scene improves reconstruction on both 7-Scenes and NRGBD.

4.3. System Performance

Table 4 reports wall-clock time and peak GPU memory on a single A100 (80 GB) for 512×384 resolution images.

At 32 views, Fast3R completes in 0.509 s using 13.25 GiB, while DUSt3R requires 129.0 s and 67.61 GiB. DUSt3R runs out of memory (OOM) at 48 views due to the quadratic memory cost of its N^2 pairwise passes. Fast3R scales smoothly to 1500 views (308.85 s, 78.59 GiB) on the same hardware.

At 1000 views, Fast3R requires 137.62 s and 63.01 GiB on a single GPU—a previously infeasible scale for end-to-end learning-based reconstruction.

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Methods	CO3Dv2 [19]					RealEstate10K [42]	FPS
	RRA@15↑	RRA@5↑	RTA@15↑	RTA@5↑	mAA(30)↑	mAA(30)↑	
Colmap+SG [6, 20]	36.1	24.4	27.3	17.2	25.3	45.2	0.056
PixSfM [14]	33.7	26.1	32.9	17.6	30.1	49.4	-
RelPose [40]	57.1	-	-	-	-	-	0.02
PosReg [31]	53.2	-	49.1	-	45.0	-	0.015
PoseDiff [31]	80.5	59.5	79.8	61.7	66.5	48.0	0.015
RelPose++ [13]	(85.5)	-	-	-	-	-	0.02
DUSt3R [32]	96.2	-	86.8	-	76.7	67.7	0.78
MASt3R [10]	94.6	93.2	**91.9**	**86.2**	81.8	**76.4**	0.23
Fast3R-no-outdoor (Ours)	**99.7**	**97.4**	87.1	76.1	**82.5**	-	**251.1**
Fast3R (Ours)	96.2	90.2	81.6	68.2	75.0	72.7	**251.1**

Table 1. **Multi-view pose regression on the CO3D [19] and RealEstate10K [42] datasets**. Parentheses denote methods that do not report results on the 10 views set; we report their best for comparison (8 views). Fast3R does not assume known camera intrinsics.

Method	FPS	7 scenes [23]		NRGBD [2]	
		Acc↓	Comp↓	Acc↓	Comp↓
F-Recon [35]	<0.1	7.62	2.31	20.59	6.31
DUSt3R† [33]	0.78	**1.23**	0.91	**2.51**	1.03
Spann3R [29]	65.4	1.48	**0.85**	3.15	1.10
Fast3R (Ours)	**251.1**	1.58	0.93	3.40	**1.01**

Table 2. **Quantitative reconstruction results on scene datasets:** The numbers indicate `median` distance to GT points on 7-Scenes [23] and NRGBD [2] datasets. These datasets contain video trajectories of 500-1500 frames, and we evaluate using the same frame skip as other baselines. For 7-Scenes this is `skip`=20, and NRGBD uses `skip`=40. DUSt3R† indicates using DUSt3R's final weights on 224 × 224 images, to fit within GPU memory. Distances are scaled 100× to remove the leading `0.00`.

Method	Views	Acc↓	Comp↓
		Med.	Med.
DUSt3R [33]	all/5	**1.159**	0.914
DUSt3R† [33]	all/5	1.297	1.002
Spann3R [29]	all/5	2.268	1.295
Fast3R (Ours)	all/5	1.706	**0.857**

Table 3. **Quantitative results on object-centric DTU [1] dataset.** Using a `skip`=5 on trajectories of 49 frames.

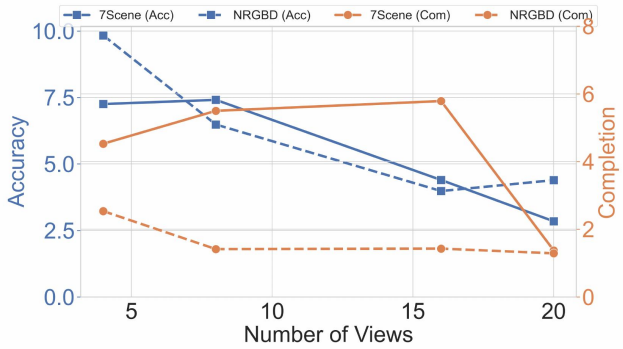

Figure 6. **Increasing # views during training: reconstruction on 7-Scenes and NRGBD.** More views during training leads to better reconstruction quality.

# Views	Fast3R		DUSt3R	
	Time (s)	Peak GPU Mem (GiB)	Time (s)	Peak GPU Mem (GiB)
2	0.065	3.84	0.092	3.52
8	0.122	6.33	8.386	24.59
32	0.509	13.25	129.0	67.61
48	0.84	20.8	OOM	OOM
320	15.938	41.90	OOM	OOM
800	89.569	55.97	OOM	OOM
1000	137.62	63.01	OOM	OOM
1500	308.85	78.59	OOM	OOM

Table 4. **System performance metrics for different view counts on Fast3R and DUSt3R on a single A100.** Time is measured in seconds (s), and memory is measured in gibibytes (GiB). Each view is 512x384 in resolution. For DUSt3R, at 48 views the N^2 pairwise reconstructions eventually consume all VRAM at its global alignment stage. Note that Fast3R's reported fastest FPS of 251.1 uses 108 views in 224x224 resolution.

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5. Ablation Studies

5.1. Effect of the Local Head

Table 5 compares using the local pointmap aligned to the global frame versus using the global pointmap directly for reconstruction. On 7-Scenes, the local-aligned approach reduces Acc from 4.81 to 2.84 and Comp from 1.64 to 1.37—significant improvements. On NRGBD, the gains are smaller but consistent. On DTU, the two approaches are nearly equivalent.

This result motivates our dual-head design: the local head's view-centric supervision makes it more accurate in local geometry, and aligning it to the global frame via the global head provides a consistent world-frame reconstruction.

5.2. Effect of Random Image-Index PE Sampling

Figure 8 validates our random image-index PE sampling strategy. Without random sampling (i.e., always using sequential indices during training), performance degrades significantly when testing with more views than seen during training—the model has not learned to treat image indices

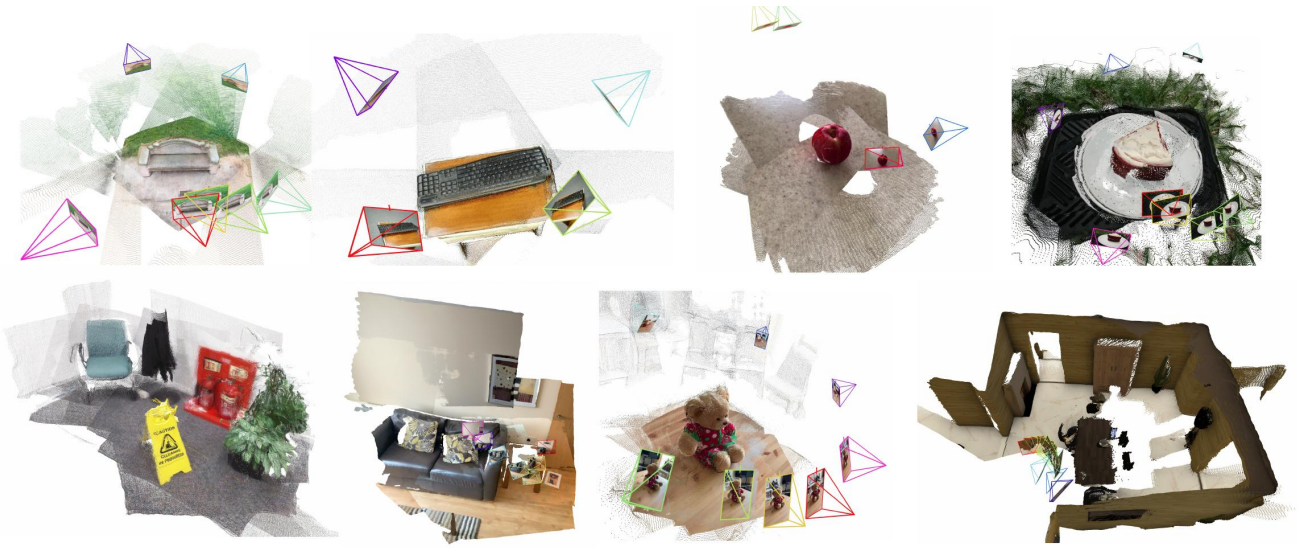

Figure 7. **Qualitative examples of Fast3R's output.** Fast3R produces high-quality 3D reconstructions and camera pose estimates from large sets of unordered, unposed images in a single forward pass.

Pointmap Type	7-Scenes [23]		NRGBD [2]		DTU	
	Acc↓	Comp↓	Acc↓	Comp↓	Acc↓	Comp↓
local aligned to global	2.84	1.37	4.39	1.28	3.91	1.41
global	4.81	1.64	4.85	1.32	3.88	1.41
Δ	+1.97	+0.27	+0.46	+0.04	-0.03	0.00

Table 5. **Ablation on the effect of local head on 3D reconstruction.** Red/green indicate an increase/decrease in error compared to using the local pointmap aligned to the global pointmap.

as arbitrary identifiers. With random sampling, Fast3R generalizes smoothly to unseen view counts, enabling "train short, test long" behavior.

6. Conclusion

We presented Fast3R, a transformer-based architecture that reconstructs 1000+ unordered, unposed images in a single forward pass. By replacing pairwise inference and global alignment with all-to-all attention over all views, Fast3R achieves near-state-of-the-art 3D reconstruction and camera pose estimation accuracy while offering orders-of-magnitude speedups. The random image-index positional embedding sampling strategy enables graceful generalization from 20 training views to 1000+ test views. A dual local/global DPT head design further improves reconstruction fidelity.

Fast3R demonstrates that scaling multi-view reconstruction to hundreds or thousands of images is not only feasible in a single forward pass, but that accuracy *improves* with more views, pointing towards a new regime of large-scale, real-time 3D perception. Looking forward, the single-pass paradigm opens promising directions: coupling Fast3R with downstream tasks such as semantic 3D scene understanding or online novel-view synthesis could unlock real-time, large-scale 3D perception from casually captured imagery.

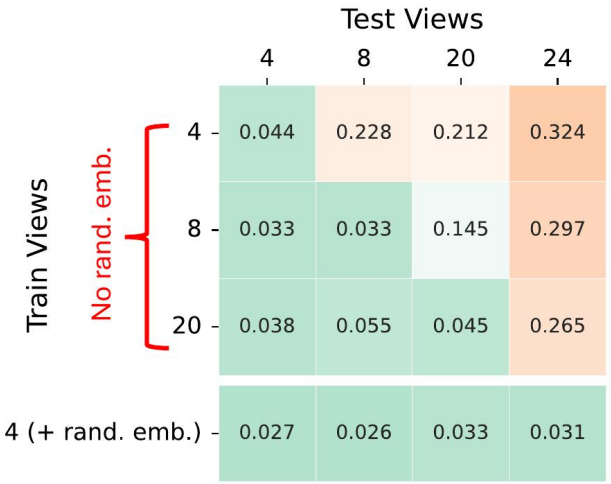

Figure 8. **Effect of sampling image-index PE during training.** Randomly sampling image indices during training (rather than using sequential indices) enables generalization to larger view counts at test time.

Limitations. Fast3R's all-to-all attention requires memory proportional to P_{total}^2, which grows with both image resolution and view count. For very large view counts (> 1000) or high-resolution inputs, memory may become a

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bottleneck. Future work may explore sparse or hierarchical attention to further extend scalability. Additionally, while Fast3R does not require known camera intrinsics, very extreme focal lengths or fisheye lenses may degrade performance. Reconstruction quality may also deteriorate in textureless or highly repetitive regions where visual correspondence is inherently ambiguous. Extending Fast3R to dynamic scenes containing moving objects remains an important direction for future work.

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